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#Number Guessing Game
#While loop & Control Statements : (else, break, continue, pass)

#Create a Python program that implements a simple number guessing game using a while loop. The program should
#statements such as else, break, and continue.

import random

# Generate a random number between 1 and 10
secret_number = random.randint(1, 10)

# Maximum attempts
attempts = 3

while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))

    # Check if the guess is out of range
    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue

    # Check the guess
    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break
    elif guess < secret_number:
        print("Too low. Try again.")
    else:
        print("Too high. Try again.")

    attempts -= 1

else:
    print("Better luck next time!")
    print("The correct number was:", secret_number)
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Guess the number (between 1 and 10): 1
Too low. Try again.
Guess the number (between 1 and 10): 2
Too low. Try again.
Guess the number (between 1 and 10): 5
Too low. Try again.
Better luck next time!
The correct number was: 7
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#For Loop:
#Multiplication Table Generator
# Get input from the user

number=int(input(" enter a number for which they want to generate a multiplication table"))
for i in range(1,11):
    total=number*i
    print(number,"x",i,"=",total)
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enter a number for which they want to generate a multiplication table5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
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#Function:
#BMI Calculator
#Create a Python program that calculates the Body Mass Index (BMI).
#Hint: BMI = weight (kg) / [height (m)]2

# Define the function
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def calculate_BMI(weight, height):  
    BMI = weight / (height ** 2)  
    return BMI  
  
# Get input from the user  
weight = float(input("Enter your weight in kg: "))  
height = float(input("Enter your height in meters: "))  
  
# Call the function  
BMI = calculate_BMI(weight, height)  
  
# Display the result  
print("Your BMI is:", format(BMI, ".2f"))
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Enter your weight in kg: 58  
Enter your height in meters: 1.62  
Your BMI is: 22.10
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